Tutorial: Create and run your first GitLab CI/CD pipeline

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This tutorial shows you how to configure and run your first CI/CD pipeline in GitLab.

If you are already familiar with basic CI/CD concepts, you can learn about common keywords in Tutorial: Create a complex pipeline.

Prerequisites

Before you start, make sure you have:

- A project in GitLab that you would like to use CI/CD for.
- The Maintainer or Owner role for the project.

If you don't have a project, you can create a public project for free on https://gitlab.com.

Steps

To create and run your first pipeline:

1. Ensure you have runners available to run your jobs.
   If you're using GitLab.com, you can skip this step. GitLab.com provides instance runners for you.
2. Create a `.gitlab-ci.yml` file at the root of your repository. This file is where you define the CI/CD jobs.

When you commit the file to your repository, the runner runs your jobs. The job results are displayed in a pipeline.

Ensure you have runners available

In GitLab, runners are agents that run your CI/CD jobs.

If you're using GitLab.com, you can skip this step. GitLab.com provides instance runners for you.

To view available runners:

1. In the top bar, select **Search or go to** and find your project.
2. In the left sidebar, select **Settings** > **CI/CD**.
3. Expand **Runners**.

As long as you have at least one runner that's active, with a green circle next to it, you have a runner available to process your jobs.

If you don't have access to these settings, contact your GitLab administrator.

If you don't have a runner

If you don't have a runner:

1. Install GitLab Runner on your local machine.

Create a `.gitlab-ci.yml` file

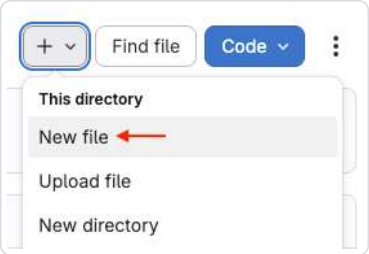
Now create a `.gitlab-ci.yml` file. It is a YAML file where you specify instructions for GitLab CI/CD.

In this file, you define:

- The structure and order of jobs that the runner should execute.
- The decisions the runner should make when specific conditions are encountered.

To create a `.gitlab-ci.yml` file in your project:

1. In the top bar, select **Search or go to** and find your project.
2. In the left sidebar, select **Code** > **Repository**.
3. Above the file list, select the branch you want to commit to. If you're not sure, leave `master` or `main`. Then, in the upper-right corner, select the plus icon (+) and **New file**:



4. For the **Filename**, type `.gitlab-ci.yml` and in the larger window, paste this sample code:

```yaml
build-job:
  stage: build
  script:
    - echo "Hello, $GITLAB_USER_LOGIN!"

test-job1:
  stage: test
  script:
    - echo "This job tests something"

test-job2:
  stage: test
  script:
    - echo "This job tests something, but takes more time than test-job1."
    - echo "After the echo commands complete, it runs the sleep command for 20 seconds"
    - echo "which simulates a test that runs 20 seconds longer than test-job1"
    - sleep 20

deploy-prod:
  stage: deploy
  script:
    - echo "This job deploys something from the $CI_COMMIT_BRANCH branch."
  environment: production
```

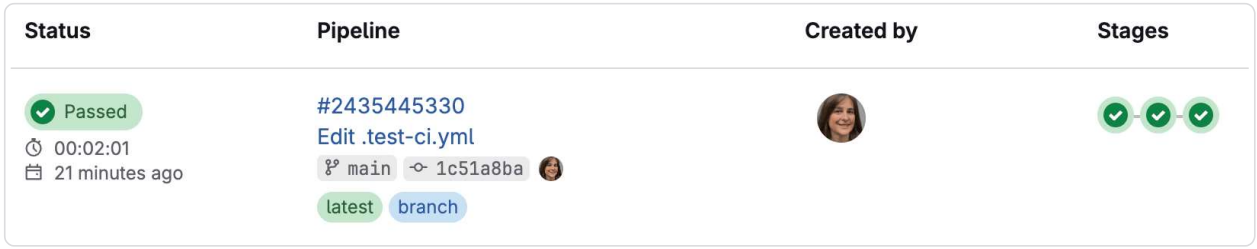
   This example shows four jobs: `build-job`, `test-job1`, `test-job2`, and `deploy-prod`. The comments listed in the `echo` commands are displayed in the UI when you view the jobs. The values for the predefined variables `$GITLAB_USER_LOGIN` and `$CI_COMMIT_BRANCH` are populated when the jobs run.

5. Select **Commit changes**.

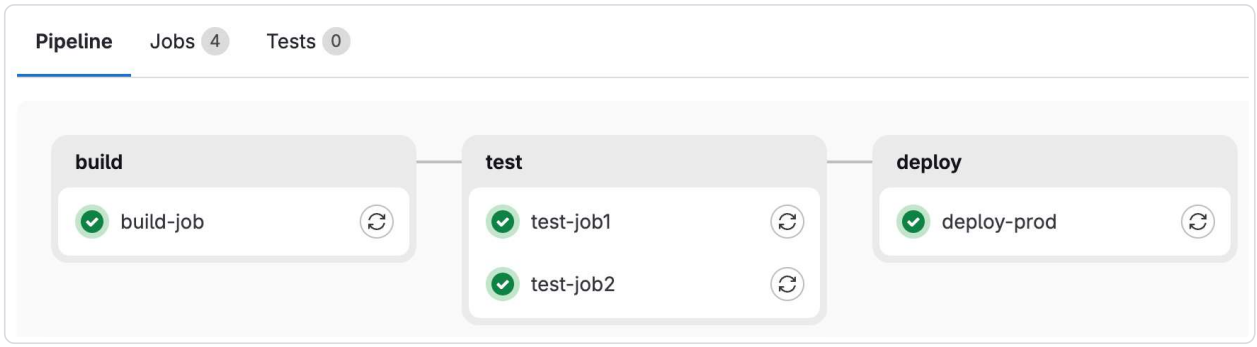
The pipeline starts and runs the jobs you defined in the `.gitlab-ci.yml` file.

Now take a look at your pipeline and the jobs within.

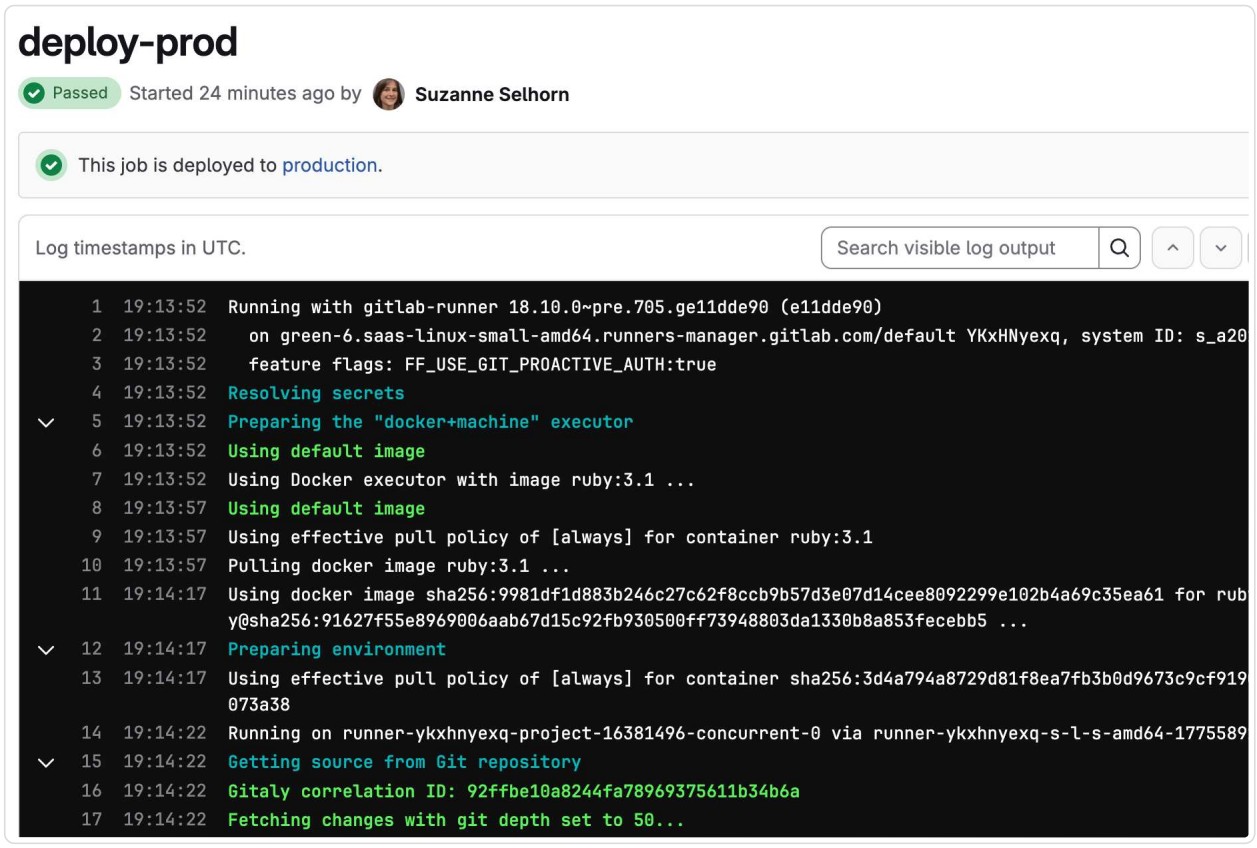
1. Go to **Build** > **Pipelines**. A pipeline with three stages should be displayed:



2. View a visual representation of your pipeline by selecting the pipeline ID (`#676` in this example):



3. View details of a job by selecting the job name. For example, `deploy-prod`:



You have successfully created your first CI/CD pipeline in GitLab. Congratulations!

Now you can get started customizing your `.gitlab-ci.yml` and defining more advanced jobs.

`.gitlab-ci.yml` tips

Here are some tips to get started working with the `.gitlab-ci.yml` file.

- Each job contains a script section and belongs to a stage:
  - `stage` describes the sequential execution of jobs. If there are runners available, jobs in a single stage run in parallel.
  - Use the `needs` keyword to run jobs out of stage order, to increase pipeline speed and efficiency.
- You can set additional configuration to customize how your jobs and stages perform:
  - Use the `rules` keyword to specify when to run or skip jobs. The `only` and `except` legacy keywords are still supported, but can't be used with `rules` in the same job.
  - Keep information across jobs and stages persistent in a pipeline with `cache` and `artifacts`. These keywords are ways to store dependencies and job output, even when using ephemeral runners for each job.
  - Use the `default` keyword to specify additional configurations that are applied to all jobs. This keyword is often used to define `before_script` and `after_script` sections that should run on every job.

Related topics

Migrate from:

- Bamboo
- CircleCI
- GitHub Actions
- Jenkins
- TeamCity

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